Linear Statistical Analysis 1

Homework 1

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Contents

Question 1 [8] **1**

Question 2 [12] **3**

Question 3 [12] **6**

Question 4 [8] **9**

Question 1 [8]

Consider the simple linear regression model

$$Y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad i = 1, \ldots, 5$$

with the following observed data

i	x_i	y_i
1	0	2
2	1	4
3	2	5
4	3	8
5	4	9

A.) What is the design matrix?
B.) What is the dimension/rank of:
- i.) $\mathbf{Y}$
- ii.) $\mathbf{X}$
- iii.) β
- iv.) rank($\mathbf{X}$)

C.) What is $\mathbf{X}^\intercal\mathbf{X}$ and $\mathbf{X}^\intercal\mathbf{Y}$?
D.) Assume $\mathbb{E}[\epsilon] = 0$ and $\text{Var}(\epsilon) = \sigma^2 I_5$. Evaluate the truthfulness of:
- i.) $\mathbb{E}[\mathbf{Y}] = \mathbf{X}\beta$
- ii.) $\text{Var}(\mathbf{Y}) = \sigma^2 I_5$
- iii.) $\mathbb{E}[\mathbf{Y}] = 0$
- iv.) The five errors all have variance σ^2

Ans:

A.) The design matrix is given by

$$\mathbf{X} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix}$$

B.)

i.) $\mathbf{Y} \in \mathbb{R}^{5\times 1}$ with rank $= 1$

ii.) $\mathbf{X} \in \mathbb{R}^{5\times 2}$ with rank $= 2$

iii.) $\beta \in \mathbb{R}^{2\times 1}$ with rank $= 1$

iv.) $\text{rank}(\mathbf{X}) \in \mathbb{R}^{1}$ and is a scalar

C.)

$$\mathbf{X}^\intercal\mathbf{X} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} 5 & 10 \\ 10 & 30 \end{bmatrix}$$

$$\mathbf{X}^\intercal\mathbf{Y} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \\ 5 \\ 8 \\ 9 \end{bmatrix} = \begin{bmatrix} 28 \\ 74 \end{bmatrix}$$

D.)

i.) True

Since $\mathbf{Y} = \mathbf{X}\beta + \epsilon$, we can take the expectation of both sides resulting in

$$\begin{aligned} \mathbb{E}[\mathbf{Y}] &= \mathbb{E}[\mathbf{X}\beta + \epsilon] \\ \mathbb{E}[\mathbf{Y}] &= \mathbb{E}[\mathbf{X}\beta] + \mathbb{E}[\epsilon] \\ \mathbb{E}[\mathbf{Y}] &= \mathbb{E}[\mathbf{X}\beta] + 0 \\ \mathbb{E}[\mathbf{Y}] &= \mathbf{X}\beta \quad \text{since} \quad \mathbb{E}[\mathbf{X}\beta] = \mathbf{X}\beta \end{aligned}$$

ii.) True

$$\begin{aligned} \text{Var}(\mathbf{Y}) &= \text{Var}(\mathbf{X}\beta + \epsilon) \\ \text{Var}(\mathbf{Y}) &= \text{Var}(\epsilon) \quad \text{since} \quad \text{Var}(\mathbf{X}\beta) = 0 \\ \text{Var}(\mathbf{Y}) &= \sigma^2 I_5 \quad \text{since} \quad \text{Var}(\epsilon) = \sigma^2 I_5 \end{aligned}$$

iii.) Usually False

Since we showed that $\mathbb{E}[\mathbf{Y}] = \mathbf{X}\beta$, this is only true in the case that $\mathbf{X}\beta = 0$ which comes from either the model predicting a constant output at 0 or if the model has only weights of 0.

iv.) True

Let ϵ have the form

$$\epsilon = \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \epsilon_3 \\ \epsilon_4 \\ \epsilon_5 \end{bmatrix}$$

then, we know that the $\text{Var}(\epsilon) = \sigma^2 I_5$ or,

$$\text{Var}(\epsilon) = \text{Var}\left(\begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \epsilon_3 \\ \epsilon_4 \\ \epsilon_5 \end{bmatrix}\right) = \begin{bmatrix} \sigma^2 & 0 & 0 & 0 & 0 \\ 0 & \sigma^2 & 0 & 0 & 0 \\ 0 & 0 & \sigma^2 & 0 & 0 \\ 0 & 0 & 0 & \sigma^2 & 0 \\ 0 & 0 & 0 & 0 & \sigma^2 \end{bmatrix}$$

so each error has the same variance of exactly σ^2.

□

Question 2 [12]

Continue using the data and design matrix from question 1.

A.) Starting from

$$S(\beta) = (\mathbf{Y} - \mathbf{X}\beta)^\intercal(\mathbf{Y} - \mathbf{X}\beta),$$

derive the normal equations

$$\mathbf{X}^\intercal\mathbf{X}\hat{\beta} = \mathbf{X}^\intercal\mathbf{Y}$$

and then state the OLS when $\mathbf{X}$ has full column rank.

B.) Using the quantities from question 1, compute

$$\hat{\beta} = \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{bmatrix}$$

Identify the values of $\hat{\beta}_0$ and $\hat{\beta}_1$, then write out the estimator $\hat{Y}$.

C.) For the least squares residual vector, $\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}}$, evaluate the truthfulness of each expression and justify geometrically:

i.) $\mathbf{Xe} = 0$
ii.) $\mathbf{X}^\intercal\mathbf{e} = 0$
iii.) $\mathbf{e}^\intercal\mathbf{e} = 0$
iv.) $\mathbf{He} = \mathbf{e}$

Ans:

a.) Given $S(\beta) = (\mathbf{Y} - \mathbf{X}\beta)^\intercal(\mathbf{Y} - \mathbf{X}\beta)$ then by expansion,

$$S(\beta) = \mathbf{Y}^\intercal\mathbf{Y} - \mathbf{Y}^\intercal\mathbf{X}\beta - \beta^\intercal\mathbf{X}^\intercal\mathbf{Y} + \beta^\intercal\mathbf{X}^\intercal\mathbf{X}\beta$$

The term $\mathbf{Y}^\intercal\mathbf{X}\beta$ has terms $\mathbf{Y}^\intercal \in \mathbb{R}^{1\times n}$, $\mathbf{X} \in \mathbb{R}^{n\times p}$, and $\beta \in \mathbb{R}^{p\times 1}$ which results in the product being in $\mathbb{R}^{1\times 1} \subseteq \mathbb{R}$. Thus,

$$\mathbf{Y}^\intercal\mathbf{X}\beta = (\mathbf{Y}^\intercal\mathbf{X}\beta)^\intercal = \beta^\intercal\mathbf{X}^\intercal\mathbf{Y}$$

Together, this simplifies $S(\beta)$ into,

$$S(\beta) = \mathbf{Y}^\intercal\mathbf{Y} - 2\beta^\intercal\mathbf{X}^\intercal\mathbf{Y} + \beta^\intercal\mathbf{X}^\intercal\mathbf{X}\beta$$

Since we are seeking the minimum value of $S(\beta)$ given β, we can take the partial derivative of $S(\beta)$ with respect to β. We can use the identity that was blatantly stolen from Wikipedia,

$$\frac{\partial(\mathbf{Ax}+\mathbf{b})^\intercal\mathbf{C}(\mathbf{Dx}+\mathbf{e})}{\partial\mathbf{x}} = \mathbf{D}^\intercal\mathbf{C}^\intercal(\mathbf{Ax}+\mathbf{b}) + \mathbf{A}^\intercal\mathbf{C}(\mathbf{Dx}+\mathbf{e})$$

with $\mathbf{b} = \mathbf{e} = 0$, $\mathbf{A} = \mathbf{D} = \mathbf{I}$, and $\mathbf{C} = \mathbf{X}^\intercal\mathbf{X}$, we get,

$$\frac{\partial S(\beta)}{\partial\beta} = -2\mathbf{X}^\intercal\mathbf{Y} + 2\mathbf{X}^\intercal\mathbf{X}\beta$$

The minimum, is of course when the derivative is equal to 0, and thus,

$$-2\mathbf{X}^\intercal\mathbf{Y} + 2\mathbf{X}^\intercal\mathbf{X}\hat{\beta} = 0$$

which leaves the final normal equations of,

$$\mathbf{X}^\intercal\mathbf{Y} = \mathbf{X}^\intercal\mathbf{X}\hat{\beta}$$

The OLS estimator is thus,

$$\hat{\beta} = (\mathbf{X}^\intercal\mathbf{X})^{-1}\mathbf{X}^\intercal\mathbf{Y}$$

b.) To find the value of $\hat{\beta}$, we can set $\hat{\beta} = (\mathbf{X}^\intercal\mathbf{X})^{-1}\mathbf{X}^\intercal\mathbf{Y}$. Copying the above from question 1, we have,

$$\mathbf{X}^\intercal\mathbf{X} = \begin{bmatrix} 5 & 10 \\ 10 & 30 \end{bmatrix}$$
$$\mathbf{X}^\intercal\mathbf{Y} = \begin{bmatrix} 28 \\ 74 \end{bmatrix}$$

Thus, it is easy to compute the inverse of $\mathbf{X}^\intercal\mathbf{X}$, assuming it exists, and can be done as,

$$(\mathbf{X}^\intercal\mathbf{X})^{-1} = \frac{1}{150-100}\begin{bmatrix} 30 & -10 \\ -10 & 5 \end{bmatrix} = \begin{bmatrix} \frac{3}{5} & -\frac{1}{5} \\ -\frac{1}{5} & \frac{1}{10} \end{bmatrix}$$

Thus, $\hat{\beta}$ is given by,

$$\begin{aligned}\hat{\beta} &= (\mathbf{X}^\intercal\mathbf{X})^{-1}\mathbf{X}^\intercal\mathbf{Y} \\ &= \begin{bmatrix} \frac{3}{5} & -\frac{1}{5} \\ -\frac{1}{5} & \frac{1}{10} \end{bmatrix}\begin{bmatrix} 28 \\ 74 \end{bmatrix} \\ &= \begin{bmatrix} 2 \\ \frac{9}{5} \end{bmatrix}\end{aligned}$$

Using the found value of $\hat{\beta}$, we can say that the estimator is thus,

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\beta}$$
$$\hat{\mathbf{Y}} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix}\begin{bmatrix} 2 \\ \frac{9}{5} \end{bmatrix}$$
$$\hat{\mathbf{Y}} = \begin{bmatrix} 2 \\ 3.8 \\ 5.6 \\ 7.4 \\ 9.2 \end{bmatrix}$$

The final estimator is thus,

$$\hat{\mathbf{Y}} = 2 + \frac{9}{5}\mathbf{X}$$

c.) The error vector $\mathbf{e}$ is thus,

$$\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}}$$
$$\mathbf{e} = \begin{bmatrix} 2 \\ 4 \\ 5 \\ 8 \\ 9 \end{bmatrix} - \begin{bmatrix} 2 \\ 3.8 \\ 5.6 \\ 7.4 \\ 9.2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0.2 \\ -0.6 \\ 0.6 \\ -0.2 \end{bmatrix}$$

i.) False

The product $\mathbf{Xe}$ is meaningless as $\mathbf{X} \in \mathbb{R}^{n\times 2}$ and $\mathbf{e} \in \mathbb{R}^{n\times 1}$. They do not have a meaningful product via usual matrix multiplication.

ii.) True

This will at least have a product which exists. Computing this, we get,

$$\mathbf{X}^\intercal\mathbf{e} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 0 \\ 0.2 \\ -0.6 \\ 0.6 \\ -0.2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Geometrically, the fitted model is the best that can be achieved. There is still some residual and the amount left is not inside the span of $\mathbf{X}$. The residual is orthogonal to the span of $\mathbf{X}$ thus leaving the inner product 0.

iii.) False

This is the magnitude of the residual vector. This can be equal to 0 but is not usually. It is only equal to 0 if the actual data is collinear.

iv.) False

With $\mathbf{H}$ defined as,

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^\intercal\mathbf{X})^{-1}\mathbf{X}^\intercal$$

We can right multiply by $\mathbf{e}$ to get,

$$\mathbf{He} = \mathbf{X}(\mathbf{X}^\intercal\mathbf{X})^{-1}\mathbf{X}^\intercal\mathbf{e}$$
$$\mathbf{He} = \mathbf{X}(\mathbf{X}^\intercal\mathbf{X})^{-1}(\mathbf{X}^\intercal\mathbf{e})$$
$$\mathbf{He} = \mathbf{X}(\mathbf{X}^\intercal\mathbf{X})^{-1}0$$
$$\mathbf{He} = 0$$

Since $\mathbf{H}$ depends only on $\mathbf{X}$, and with $\mathbf{e}$ outside the span of $\mathbf{X}$, how could $\mathbf{H}$ ever reach a point that is orthogonal to it? It can't leaving the identity as false.

□

Question 3 [12]

Continue using the data and fitted regression model from before.

a.) Compute both $\hat{\mathbf{Y}} = \mathbf{X}\hat{\beta}$ and $\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}}$.
b.) Determine the truth value of:
i.) The residuals satisfy

$$\sum_{i=1}^{n} \mathbf{e}_i = 0$$

ii.) The fitted values satisfy $\overline{\hat{\mathbf{Y}}} = \overline{\mathbf{Y}}$.
iii.) The vector composing of all 1's (i.e. $\mathbf{1}_n$) belongs to $\mathcal{C}(\mathbf{X})$.
iv.) These properties necessarily hold for every regression model without an intercept.
c.) Define $\mathbf{M} = \mathbf{I}_n - \mathbf{H}$. Determine the truth value of:
i.) $\mathbf{e} = \mathbf{M}\mathbf{Y}$
ii.) $\mathbf{M}^\intercal = \mathbf{M}$
iii.) $\mathbf{M}^2 = \mathbf{M}$
iv.) $\mathbf{H}\mathbf{M} = \mathbf{0}$
d.) Suppose now that a design matrix $\mathbf{X}$ does not have full column rank. Determine the truth value of:
i.) $(\mathbf{X}^\intercal\mathbf{X})^{-1}$ necessarily exists.
ii.) The least squares coefficient vector $\hat{\beta}$ may not be unique.
iii.) The fitted vector $\hat{\mathbf{Y}}$ is still unique.
iv.) The residual vector $\mathbf{e}$ is still unique.

Ans:

a.) Copying from above,

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\beta}$$
$$\hat{\mathbf{Y}} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ \frac{9}{5} \end{bmatrix}$$
$$\hat{\mathbf{Y}} = \begin{bmatrix} 2 \\ 3.8 \\ 5.6 \\ 7.4 \\ 9.2 \end{bmatrix}$$

and,

$$\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}}$$
$$\mathbf{e} = \begin{bmatrix} 2 \\ 4 \\ 5 \\ 8 \\ 9 \end{bmatrix} - \begin{bmatrix} 2 \\ 3.8 \\ 5.6 \\ 7.4 \\ 9.2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0.2 \\ -0.6 \\ 0.6 \\ -0.2 \end{bmatrix}$$

b.)
i.) True

Taking $\mathbf{e}$, we get,

$$\begin{bmatrix} 0 \\ 0.2 \\ -0.6 \\ 0.6 \\ -0.2 \end{bmatrix} \implies 0 + 0.2 - 0.6 + 0.6 - 0.2 = 0$$

ii.) True

$$\overline{\mathbf{Y}} = \frac{2+4+5+8+9}{5} = 5.6$$
$$\overline{\hat{\mathbf{Y}}} = \frac{2+3.8+5.6+7.4+9.2}{5} = 5.6$$

iii.) True

If we partition $\mathbf{X}$ as $\mathbf{X} = [\mathbf{x}_1 \quad \mathbf{x}_2]$, then $\mathbf{x}_1 = \mathbf{1}_5$.

iv.) False

The identity of,

$$\sum_{i=1}^{n} \mathbf{e}_i = 0$$

does not hold if the intercept drops.

c.) Here, $\mathbf{M} = \mathbf{I}_5 - \mathbf{H}$,

$$\mathbf{M} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 0.6 & 0.4 & 0.2 & 0.0 & -0.2 \\ 0.4 & 0.3 & 0.2 & 0.1 & 0.0 \\ 0.2 & 0.2 & 0.2 & 0.2 & 0.2 \\ 0.0 & 0.1 & 0.2 & 0.3 & 0.4 \\ -0.2 & 0.0 & 0.2 & 0.4 & 0.6 \end{bmatrix}$$

$$= \begin{bmatrix} 0.4 & -0.4 & -0.2 & 0.0 & 0.2 \\ -0.4 & 0.7 & -0.2 & -0.1 & 0.0 \\ -0.2 & -0.2 & 0.8 & -0.2 & -0.2 \\ 0.0 & -0.1 & -0.2 & 0.7 & -0.4 \\ 0.2 & 0.0 & -0.2 & -0.4 & 0.4 \end{bmatrix}$$

i.) True

Calculating directly,

$$\mathbf{MY} = \begin{bmatrix} 0.4 & -0.4 & -0.2 & 0.0 & 0.2 \\ -0.4 & 0.7 & -0.2 & -0.1 & 0.0 \\ -0.2 & -0.2 & 0.8 & -0.2 & -0.2 \\ 0.0 & -0.1 & -0.2 & 0.7 & -0.4 \\ 0.2 & 0.0 & -0.2 & -0.4 & 0.4 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \\ 5 \\ 8 \\ 9 \end{bmatrix}$$
$$= \begin{bmatrix} 0 \\ 0.2 \\ -0.6 \\ 0.6 \\ -0.2 \end{bmatrix}$$
$$= \mathbf{e}$$

ii.) True

Since $\mathbf{M}$ is defined as $\mathbf{M} = \mathbf{I} - \mathbf{H}$, we can use the properties of both $\mathbf{I}$ and $\mathbf{H}$.

$$\begin{aligned}\mathbf{M}^\intercal &= (\mathbf{I} - \mathbf{H})^\intercal \\ &= \mathbf{I}^\intercal - \mathbf{H}^\intercal \\ &= \mathbf{I} - \mathbf{H} \quad \text{since} \quad \mathbf{I}^\intercal = \mathbf{I} \quad \& \quad \mathbf{H}^\intercal = \mathbf{H} \\ &= \mathbf{M}\end{aligned}$$

iii.) True

$$\begin{aligned}\mathbf{M}^2 &= (\mathbf{I} - \mathbf{H})(\mathbf{I} - \mathbf{H}) \\ &= \mathbf{I}^2 - \mathbf{HI} - \mathbf{IH} + \mathbf{H}^2 \\ &= \mathbf{I} - 2\mathbf{H} + \mathbf{H} \quad \text{since} \quad \mathbf{H}^2 = \mathbf{H} \\ &= \mathbf{I} - \mathbf{H} \\ &= \mathbf{M}\end{aligned}$$

iv.) True

$$\begin{aligned}\mathbf{HM} &= \mathbf{H}(\mathbf{I} - \mathbf{H}) \\ &= \mathbf{HI} - \mathbf{H}^2 \\ &= \mathbf{H} - \mathbf{H} \quad \text{since} \quad \mathbf{H}^2 = \mathbf{H} \\ &= 0\end{aligned}$$

d.) We assume that $\mathbf{X}$ does not have full column rank.

i.) False

This is due to the following properties. Since $\mathbf{X}$ does not have full column rank, its determinant is 0. The value $(\mathbf{X}^\intercal\mathbf{X})^{-1}$ exists iff that $\det[\mathbf{X}^\intercal\mathbf{X}] \neq 0$.

$$\begin{aligned}&\text{rank}(\mathbf{X}^\intercal\mathbf{X}) = \text{rank}(\mathbf{X}) \\ &\text{rank}(\mathbf{X}) < p \implies \text{rank}(\mathbf{X}^\intercal\mathbf{X}) < p \\ &\text{rank}(\mathbf{X}^\intercal\mathbf{X}) < p \implies \det(\mathbf{X}^\intercal\mathbf{X}) = 0\end{aligned}$$

Which is a contradiction leading to False.

ii.) True

The values of β do not need to be unique.

iii.) True

Due to $\mathbf{X}$ not having full column rank, the $\hat{\beta}$ that is chosen, while not unique, does point toward a unique $\hat{\mathbf{Y}}$ which minimizes the least squares problem.

iv.) True

Taking $\hat{\mathbf{Y}}$ to be unique and $\mathbf{Y}$ to be constant, then necessarily, $\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}}$ is also unique since it is just an affine shift of $\hat{\mathbf{Y}}$.

□

Question 4 [8]

Using the built-in model in R, *cars*, consider the model,

$$\text{dist}_i = \beta_0 + \beta_1 \text{speed}_i + \epsilon_i$$

a.) Fit the model using linear regression. Identify the vector β and interpret β_1.
b.) Construct $\mathbf{Y}$ and $\mathbf{X}$ manually. Reproduce the OLS coefficients using matrix operations.

Ans:

a.) Running the code (in accompanying file), I get the β vector to be,

$$\beta = \begin{bmatrix} -17.579 \\ 3.932 \end{bmatrix}$$

This gives an overall linear regression of,

$$\widehat{\text{dist}}_i = -17.579 + 3.932 \text{speed}_i$$

What β_1 means is that the distance required to stop goes up by 3.932 feet for every mph increase of speed.

```
data(cars)
fit <- lm(dist ~ speed, data = cars)
summary(fit)
```

```
##
## Call:
## lm(formula = dist ~ speed, data = cars)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -29.069  -9.525  -2.272   9.215  43.201
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -17.5791     6.7584  -2.601   0.0123 *
## speed         3.9324     0.4155   9.464 1.49e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 15.38 on 48 degrees of freedom
## Multiple R-squared:  0.6511, Adjusted R-squared:  0.6438
## F-statistic: 89.57 on 1 and 48 DF,  p-value: 1.49e-12
```

b.) The data and targets are shown,

$$\mathbf{X} = \begin{bmatrix} 1 & 4 \\ 1 & 4 \\ 1 & 7 \\ \vdots & \vdots \\ 1 & 25 \end{bmatrix} \quad \& \quad \mathbf{Y} = \begin{bmatrix} 2 \\ 10 \\ 4 \\ \vdots \\ 85 \end{bmatrix}$$

For brevity, I am not going to write those out explicitly but I am going to use them to hand calculate the β vector. We know that $\hat{\beta}$ is given by,

$$\hat{\beta} = (\mathbf{X}^\intercal \mathbf{X})^{-1} \mathbf{X}^\intercal \mathbf{Y}$$

We can painfully do the matrix operations to get that,

$$\mathbf{X}^\intercal \mathbf{X} = \begin{bmatrix} 50 & 770 \\ 770 & 13228 \end{bmatrix} \quad \& \quad \mathbf{X}^\intercal \mathbf{Y} = \begin{bmatrix} 2149 \\ 38482 \end{bmatrix}$$

These are easier to work with. By taking the inverse we get,

$$(\mathbf{X}^\intercal \mathbf{X})^{-1} = \frac{1}{34250} \begin{bmatrix} 6614 & -385 \\ -385 & 25 \end{bmatrix}$$

We thus get the multiplication problem of,

$$\begin{aligned} \hat{\beta} &= (\mathbf{X}^\intercal \mathbf{X})^{-1} \mathbf{X}^\intercal \mathbf{Y} \\ &= \frac{1}{34250} \begin{bmatrix} 6614 & -385 \\ -385 & 25 \end{bmatrix} \begin{bmatrix} 2149 \\ 38482 \end{bmatrix} \\ &= \begin{bmatrix} -17.579 \\ 3.932 \end{bmatrix} \end{aligned}$$

Which is exactly what our R code produced.

```
Y <- matrix(cars$dist, ncol = 1)
X <- cbind(1, cars$speed)

# X^T X
XtX <- t(X) %*% X
XtX
```

```
##      [,1]  [,2]
## [1,]   50   770
## [2,]  770 13228
```

```
# X^T Y
XtY <- t(X) %*% Y
XtY
```

```
##       [,1]
## [1,]  2149
## [2,] 38482
```

```
beta_hat <- solve(t(X) %*% X) %*% t(X) %*% Y
beta_hat
```

```
##            [,1]
## [1,] -17.579095
## [2,]   3.932409
```

□